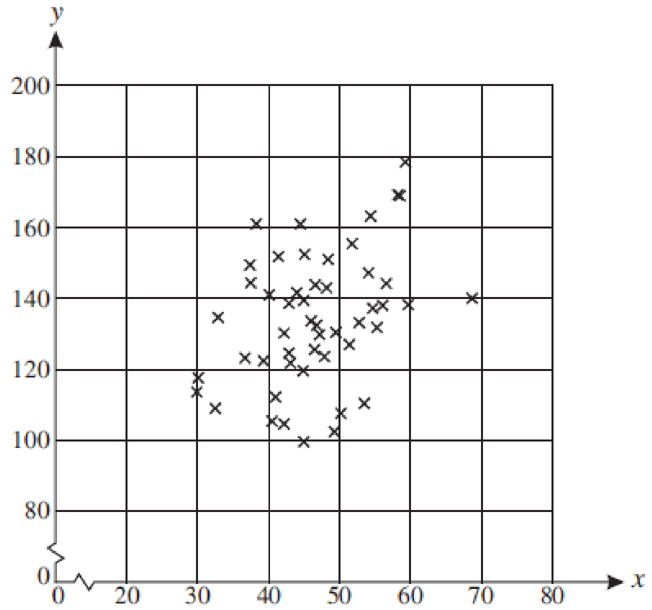
Question 2 (from MEI S2 June 2009 – edited)

An investment analyst thinks that there may be correlation between the cost of oil, x dollars per barrel, and the price of a particular share, y pence. The analyst selects 50 days at random and records the values of x and y. Summary statistics for these data are shown below, together with a scatter diagram.



The value of the product moment correlation coefficient for these data is 0.3767.

(ii) Carry out a hypothesis test at the 5% significance level to investigate the analyst's belief. State your hypotheses clearly, defining any symbols which you use. **[6]**

(iv) Explain why a 2-tail test is appropriate even though it is clear from the scatter diagram that the sample has a positive correlation coefficient. **[2]**

Question 3 (from MEI S2 June 2012 – edited)

The times taken by ten randomly selected competitors for the first and last sections of an Olympic bobsleigh run are found to have a prodcut moment correlation coefficient of 0.6770.

(ii) Carry out a hypothesis test at the 10% significance level to investigate whether there is any correlation between times taken for the first and last sections of the bobsleigh run. **[6]**

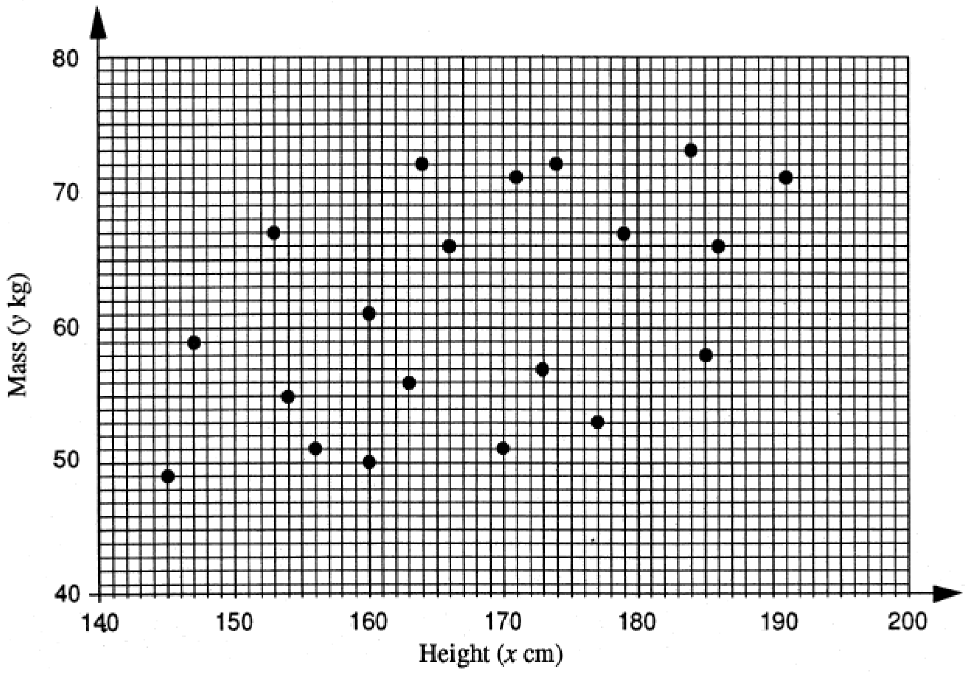
(iv) A commentator says that in order to have a fast time on the last section, you must have a fast time on the first section. Comment briefly on this suggestion. **[2]**

(v) *(A)* Would your conclusion in part **(ii)** have been different if you had carried out the hypothesis test at the 1% level rather than the 10% level? Explain your answer. **[2]**

(B) State one advantage and one disadvantage of using a 1% significance level rather than a 10% significance level in a hypothesis test. **[2]**

Question 4 (from MEI S2 June 1997 – edited)

The bivariate sample illustrated in the scatter diagram (Fig. 1) shows the heights, x cm, and masses, y kg, of a random sample of 20 students.



The value of the product moment correlation coefficient for these data is 0.4735.

(ii) Carry out a hypothesis test, at the 5% level of significance, to determine whether or not there is evidence that the height of a student is positively correlated with his or her mass. [7]

(iii) A statistics student suggests that a positive correlation between height and mass implies that "the taller a student is the heavier he or she will be". Comment on this statement with reference to your conclusions in part **(ii)**. [3]

Question 5 (from MEI S2 June 2008 – edited)

A researcher believes that there is a negative correlation between money spent by the government on education and population growth in various countries. A random sample of 48 countries is selected to investigate this belief. The level of government spending on education x, measured in suitable units, and the annual percentage population growth rate y, are recorded for these countries.

The value of the product moment correlation coefficient between x and y is found to be -0.2740.

(ii) Carry out a hypothesis test at the 5% significance level to investigate the researcher's belief. State your hypotheses clearly, defining any symbols which you use. [6]

(iv) A student suggests that if the variables are negatively correlated then population growth rates can be reduced by increasing spending on education. Explain why the student may be wrong. Discuss an alternative explanation for the correlation. [3]

(v) State briefly one advantage and one disadvantage of using a smaller sample size in this investigation. [2]